

Language Endangerment Status Analysis

1. Dataset Overview

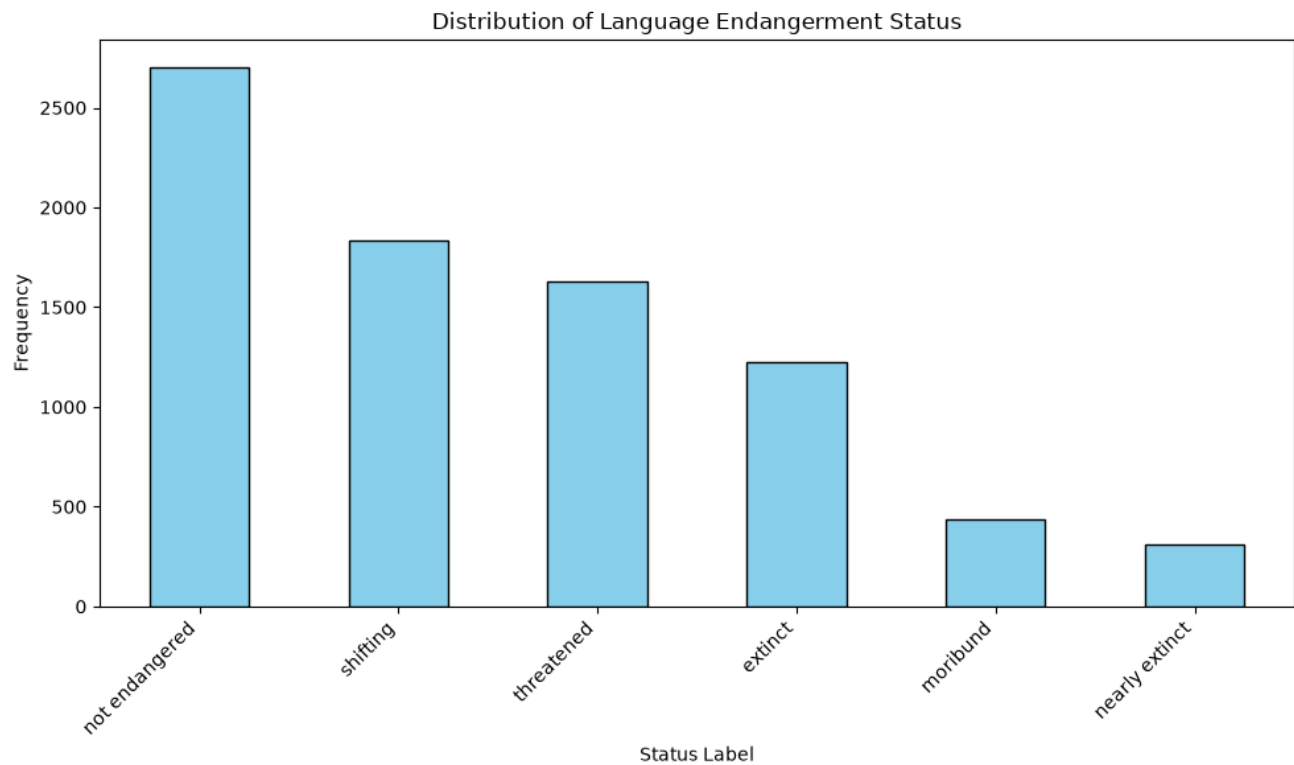
This dataset contains information on 8,612 languages worldwide, exploring their geographic locations, macroareas, linguistic families, and endangerment status. The primary goal is to understand global language vitality and distribution patterns.

2. Data Quality Notes

- **Missing Values:** `iso639p3code` has the highest missing rate (~8.8%), followed by `status_code` and `status_label` (~5.5%). `latitude` and `longitude` are missing in ~3.6% of records.
- **Completeness:** The target variables (`status_code`, `status_label`) are mostly complete, allowing for robust analysis of the ~8,100 labeled languages.
- **Data Types:** Appropriate variable types are used, including boolean flags for isolates and numerical coordinates for mapping.

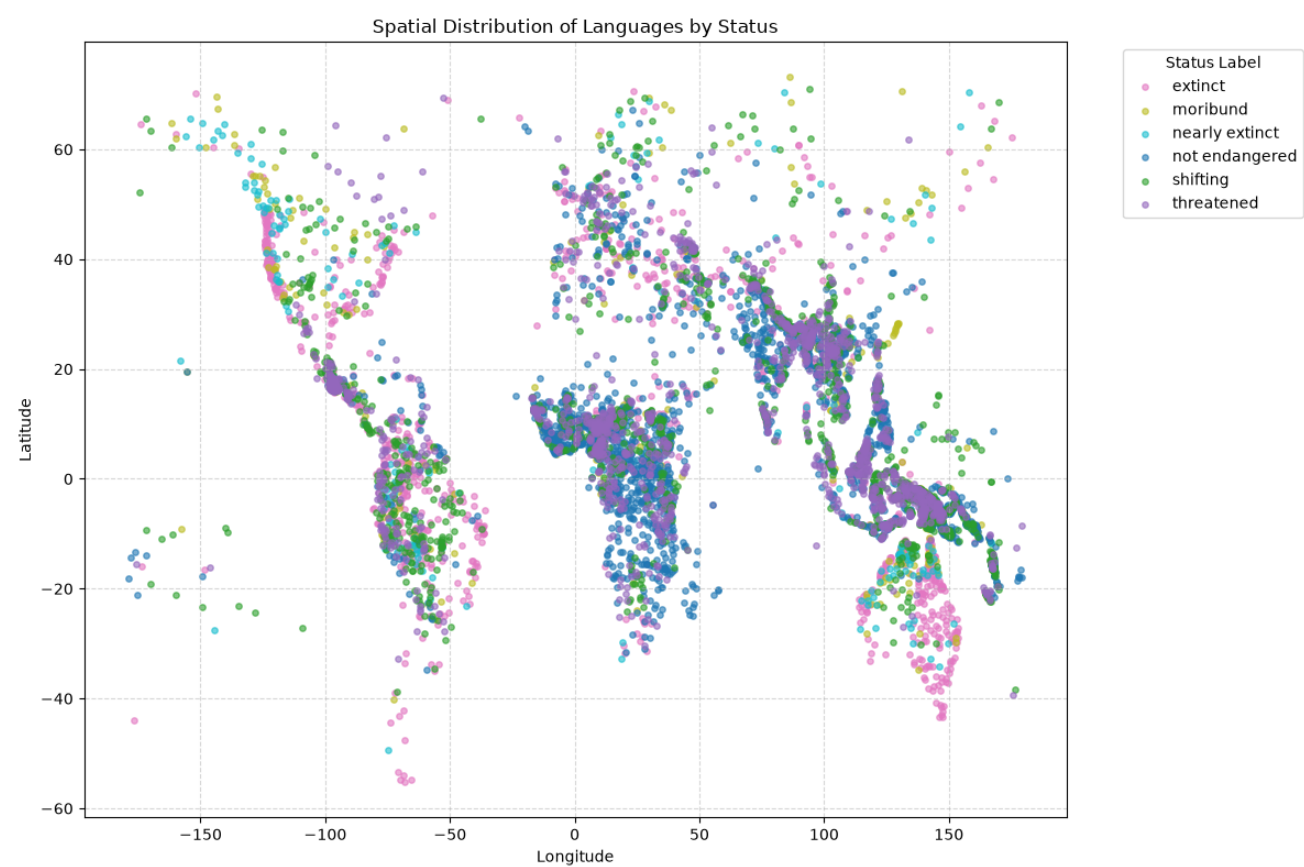
3. Exploratory Analysis

Chart 1: Language Endangerment Status Distribution



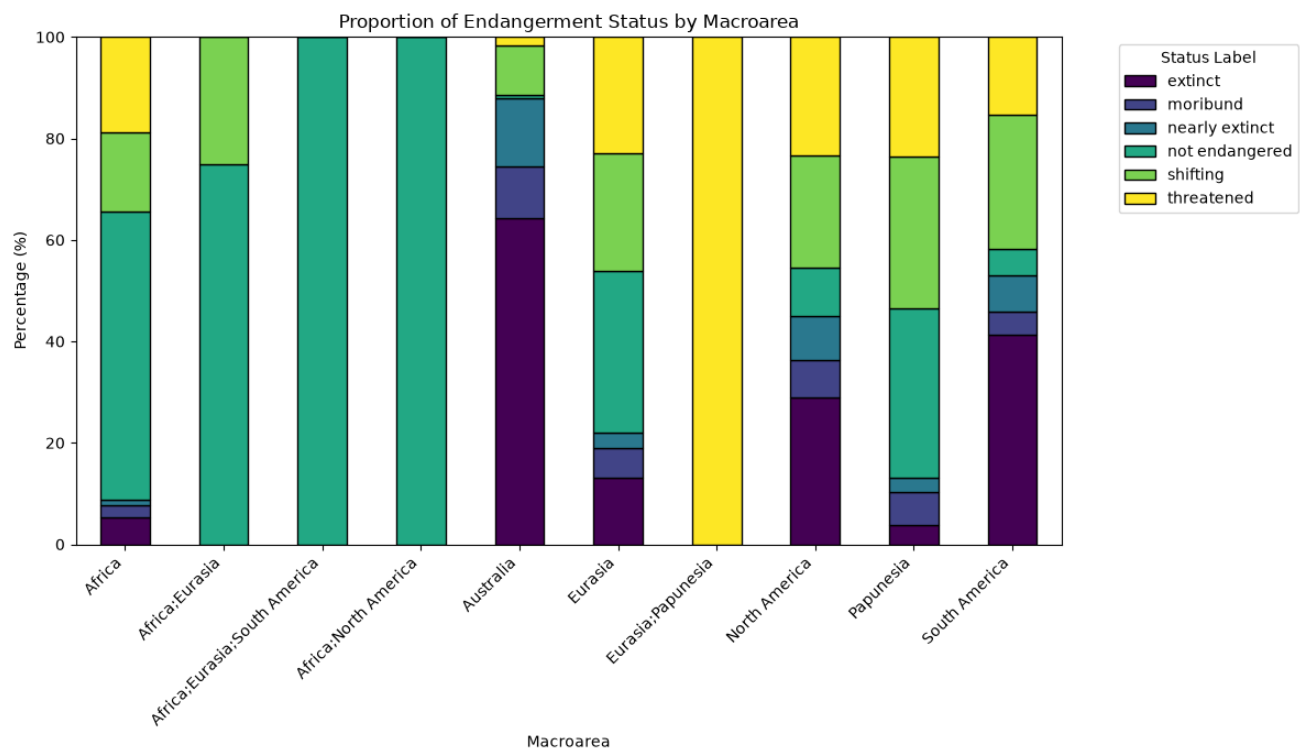
What it shows: Most languages in the dataset are classified as 'not endangered'.

Chart 2: Geographical Distribution of Languages



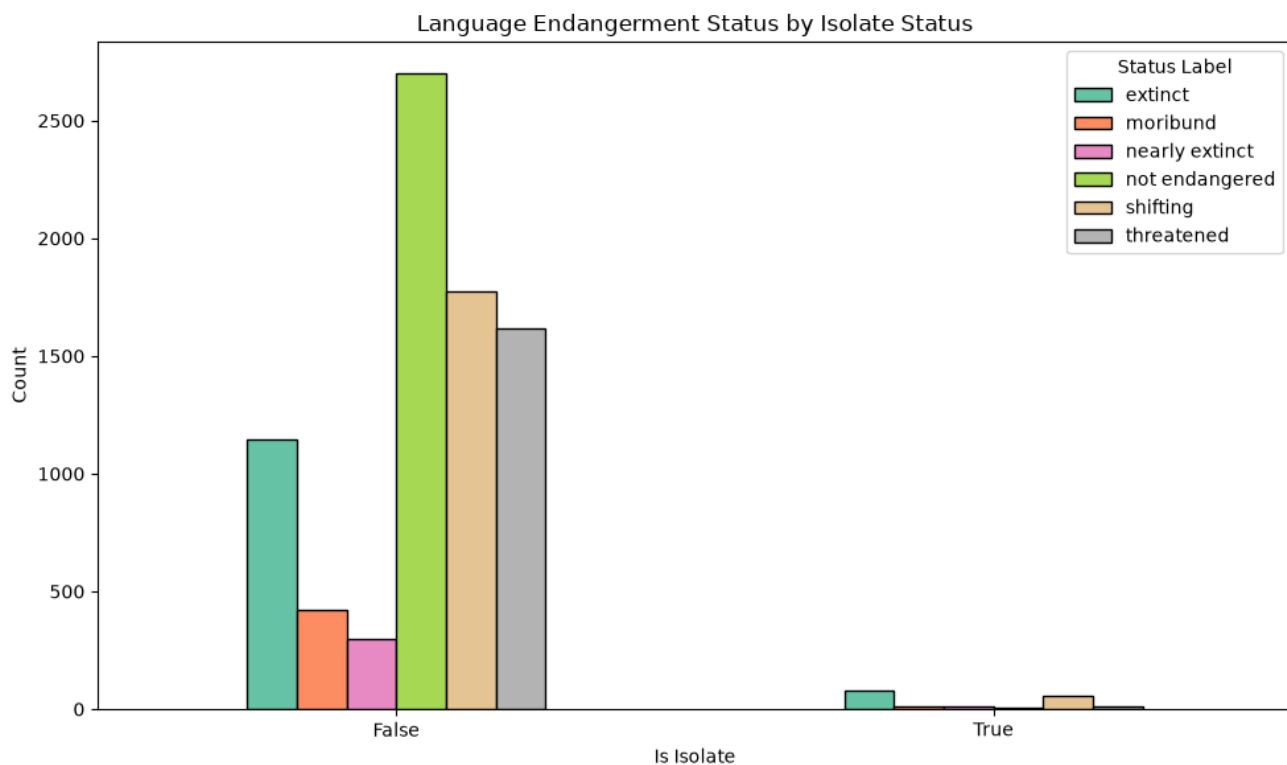
What it shows: Languages and their endangerment status show distinct regional clusters across the globe.

Chart 3: Endangerment Status Proportions per Macroarea



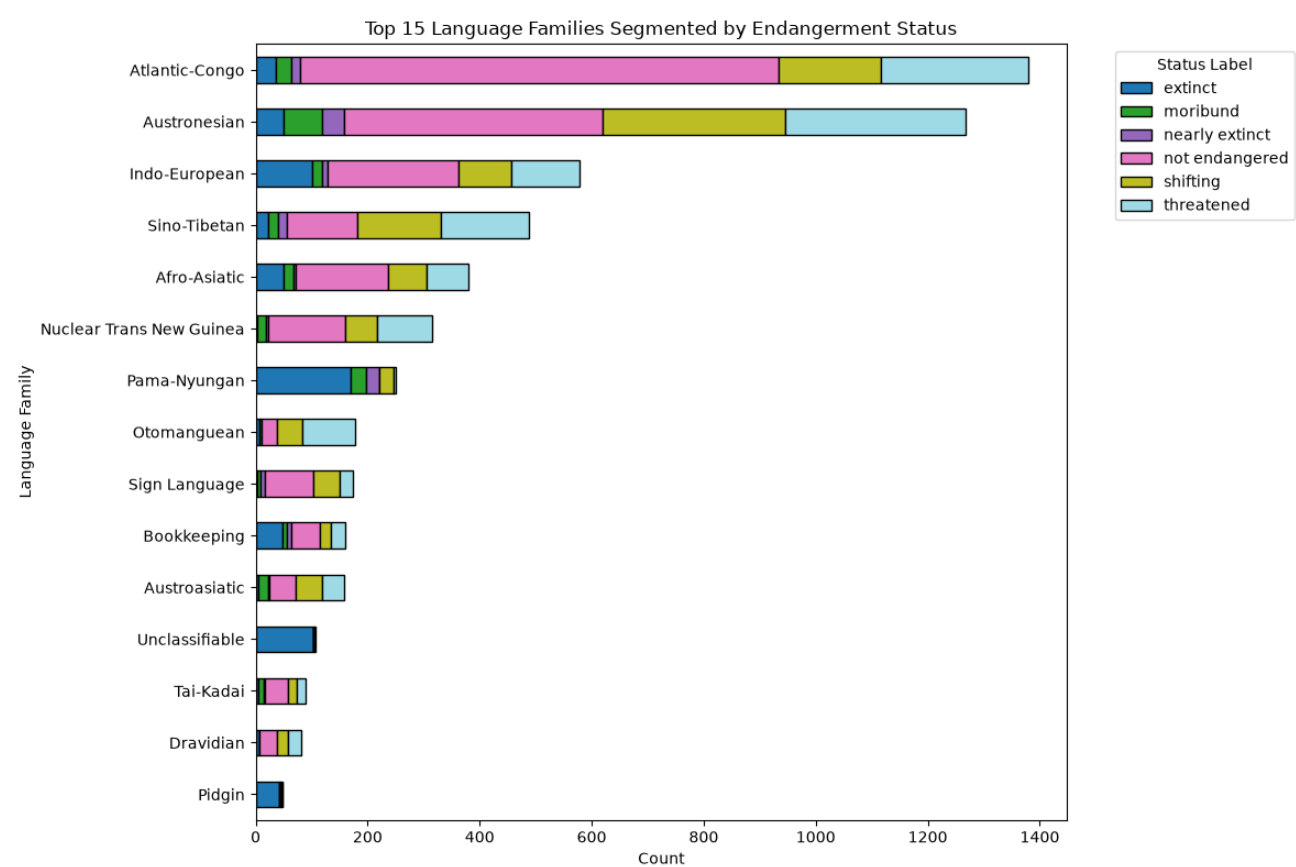
What it shows: Endangerment severity proportions vary significantly across different geographical macroareas.

Chart 4: Endangerment Status of Language Isolates vs Families



What it shows: The vast majority of languages belong to families, with isolates showing distinct endangerment profiles.

Chart 5: Top 15 Language Families by Endangerment Status



What it shows: The largest language families have varied proportions of endangered and extinct languages.

4. Modeling Results

Class	P	R	F1-Score	Support
extinct	0.0666	0.0666	0.0666	245
moribund	0.1206	0.1008	0.1108	87
nearly extinct	0.1313	0.1315	0.1315	62
not endangered	0.0172	0.0266	0.0216	541
shifting	0.4743	0.4848	0.4795	367

threatened	0.423739	326
Accuracy	0.53	1628
Macro Avg	0.414040	1628
Weighted Avg	0.515351	1628

5. Key Insights

- **Language Endangerment Status Distribution:** "Not endangered" is the most common single classification, accounting for roughly a third (2,704) of the 8,138 languages with a documented status. However, a wide spread of vulnerability exists across the dataset. Nearly 5.5% of the languages in the dataset are missing their endangerment status labels, which could slightly skew the true global distribution.
- **Geographical Distribution:** Language densities are heavily concentrated in specific global regions, most prominently reflecting areas like Africa and Papunesia. Endangerment statuses visually cluster together, suggesting that geographic proximity strongly influences a language's likelihood of survival.
- **Endangerment by Region and Family:** The proportion of safe to endangered languages varies drastically by macroarea, pointing to localized extinction crises. While major language families like Atlantic-Congo dominate the dataset, their internal distributions of endangerment vary wildly.
- **Isolates:** The overwhelming majority of documented languages belong to a language family, while isolated languages form a very small minority. Lacking broader linguistic support networks, their endangerment profiles likely reflect higher vulnerability.
- **Modeling Performance:** The predictive baseline model struggles significantly to classify intermediate endangerment stages like "moribund" (0.08 F1-score) and "nearly extinct" (0.15 F1-score), suggesting these categories possess highly overlapping geographic and linguistic features.

6. Recommendations / Next Steps

- Impute or further investigate the missing target labels (status_code/status_label) to ensure a completely accurate picture of global linguistic vitality.
- Develop specialized, localized conservation strategies specifically targeting language isolates, given their unique vulnerability and lack of familial linguistic ties.
- Explore advanced modeling techniques or additional socioeconomic features to improve the classification performance on intermediate endangerment categories (e.g., "moribund", "nearly extinct").
- Conduct deep-dive regional analyses into macroareas showing disproportionately high rates of language extinction to uncover underlying sociopolitical causes.